Aditya Bhatia
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Topic: Assignment -1 Date:

Sec A - Introduction to DBMS

Q1 DBMS → A DBMS is software that manages
Ans data in an organized way using tables. It helps users to store, update & retrieve data easily. It is required to manage large data safely, avoid duplication & provide secure access.

Q2 Why DBMS is better than file system?
Ans DBMS stores data in a structured format & allows multiple users to access it safely. File systems may create duplicate data and are harder to manage.

Q3 Data Abstraction: It means hiding technical detail and showing only required information to user.
- Physical (how data is stored)
- Logical (what data is stored)
- View (what user sees)

Q4 What is three-schema architecture?
Ans It divides database into three part:
- Internal level
- Conceptual level
- External level

It helps in data independence & easy modification.

Q5 Data Independence: It means changing dataset structures and one level without affecting other levels.

Teacher's Sign................

Q6 ER Model - It is a diagram that represents entities, their attributes & relationships it is used for database design planning.

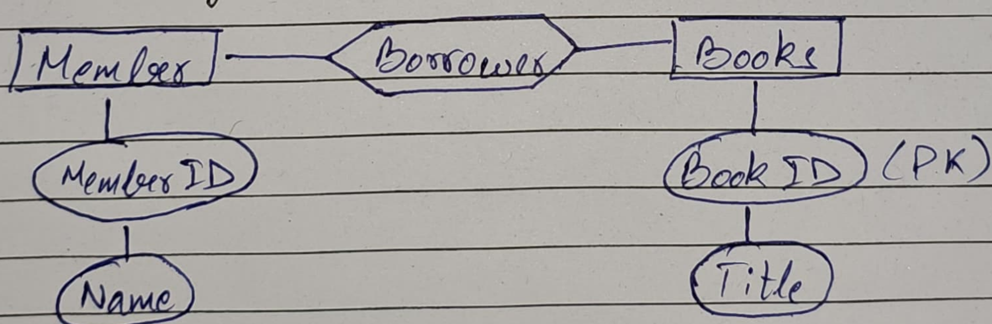
Q7 What are Integrity Constraints - Integrity constraints are rules that ensure correct & valid data.
Ex → Primary keys, foreign keys, Not Null.

Section B Design, Draw & Defend

1. Three Schema

External Level → User View
↓
Conceptual Level → Overall Structure
↓
Internal ~~Ve~~ Level → Physical Storage

2. ER Diagram → Library



→ Entities : Member, Book
→ Relationship : Borrower
→ PK : Member ID, Book ID

One Member can borrow many books

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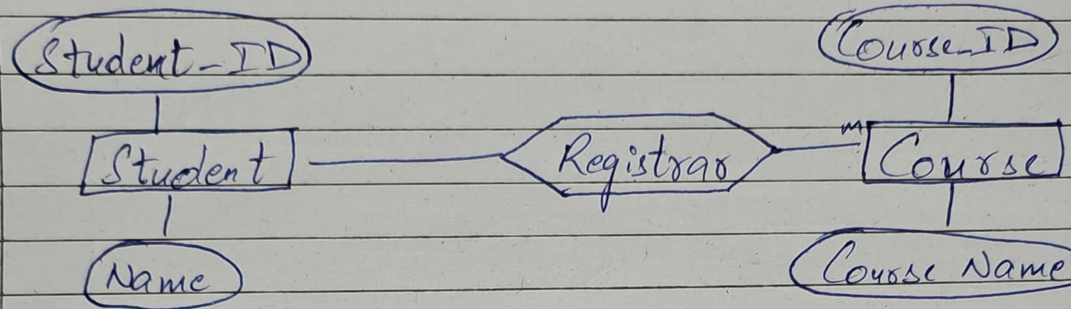
iii) Student - Course Registration ER
- Student : (Student_ID, Name)
- Course : (Course - ID, Course Name)



Section C - Case Based Analytical Q

i) ER Diagram - Hospital System

Entities : Patient, Doctor, Appointment
Patients books Appointments
· Doctor attends Appointment

Primary Keys : Patient ID, Doctor_ID,
Appointment _ ID

ii) Online Shopping :
Customer . (Customer ID_PK)
Product (Product _ID_PK)
Order (Order -ID_ PK)
foreign key in order table connect customer &
product.

iii) University -Data Abstraction
Student see marks (view level)
Admin manager records (logical level)

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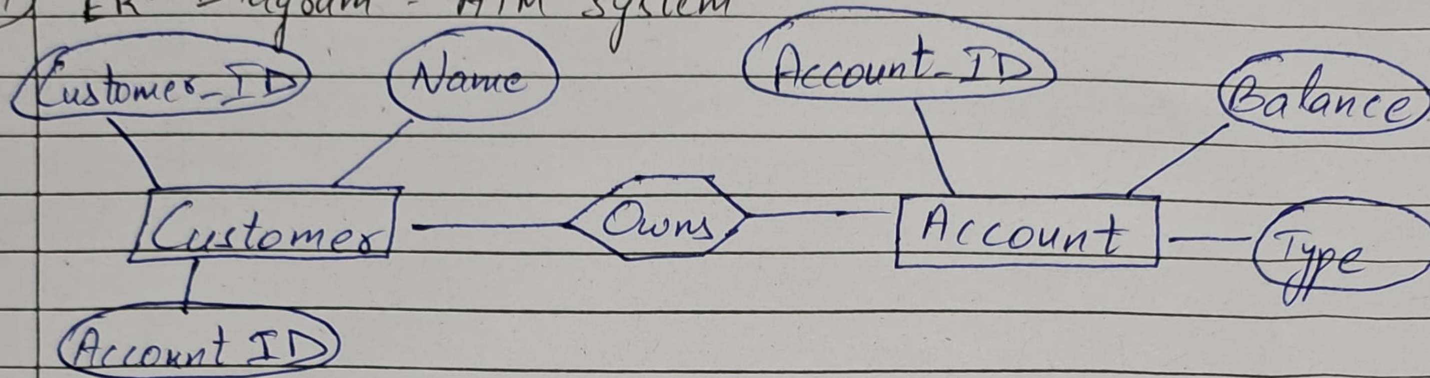
DBA . control storage (Physical level)

Section D - Onediagram + On Insight

i) ER Diagram - ATM System



Account_ID is primary key
Relation : Customer owns account

Account ID take uniquely identifies Account.

ii) Three - Schema Architecture + One Benefit
⇒ Benefit : Provides data independence
Change in storage do not affect users.

Diagram : External level (user view)
↓
Conceptual level (logical structure of Database)
↓
Internal level (Physical storage level)

iii) Draw an Entity with Attributes

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Entity: Employee
Employee_ID (PK)
Name
Salary
Department

Primary Key: EmployeeID because for all employees their ID will be different.

Section E - Reflection, Assumption & Viva

1. Two assumptions while drawing ER-diagram.
→ Each student having a unique Student_ID.
One appointment being linked to one doctor.

2. One limit of ER Modelling.
→ Cannot represent complex constraints easily.
- Does not show procedural aspects
Not suitable for large dynamic system directly.

3. Most difficult part in Unit 1?
→ The most difficult part was understanding data independence because it requires clear understanding of three-schema, architecture & how levels interact with each other.